Shashank Shukla

Senior Platform Engineer

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SUMMARY

Platform engineer with 5+ years designing and shipping production software for distributed, cloud-native systems on AWS, GCP and Kubernetes. Builds platforms end to end: a bring-your-own-cloud delivery platform that turns one config file into a full customer environment in 45 minutes, an event-driven AI incident-response service on Amazon Bedrock, a Kubernetes operator for self-hosted language models, and a FastAPI SaaS API platform. Leads and mentors a 4-engineer team, sets engineering standards, and owns reliability and performance for 270+ microservices across 12 EKS clusters.

EXPERIENCE

Nielsen Bangalore, India

Senior DevOps Engineer Jun 2025 – Present

- **Architected and built a bring-your-own-cloud (BYOC) delivery platform** that turns one customer config file into a complete production environment in the customer's AWS or GCP account: Pulumi infrastructure-as-code provisions networking, Kubernetes, databases and storage, then the platform bootstraps Argo CD and Karpenter and deploys 40-45 microservices. Cut environment setup from 6-7 hours to 45 minutes; live for 4 customers.
- **Designed the platform's release-management layer** so infrastructure and application upgrades for existing customers run through the same automated pipeline and are driven from the customer dashboard, serving both BYOC and SaaS deployments.
- **Designed and built an event-driven AI incident-response service:** Prometheus alerts from 12 EKS clusters trigger retrieval over a Bedrock Knowledge Base of runbooks and docs, LLM triage on Amazon Bedrock, automated remediation of common failures such as OOM kills, and escalation to on-call through incident.io with a diagnosis.
- **Built a custom Kubernetes CRD and controller** that runs two self-hosted open-weight small language models as a managed fleet on EKS, at ~4x lower inference cost than managed APIs.
- **Built the real-time Ads platform infrastructure from scratch to production,** designed for sub-10 ms responses: evaluated ScyllaDB and Aerospike with vendor teams and used AWS Local Zones and Global Accelerator to cut network latency.
- **Own reliability and performance for 270+ microservices on 12 EKS clusters:** zero-downtime upgrades from v1.32 to v1.34, blue-green deployments, and an observability stack on Prometheus, Loki, Grafana and incident.io; node memory swap cut OOM kills ~70%.
- **Reduced AWS spend by ~$1,800/month** by consolidating 20+ ALBs behind NGINX Ingress for 50+ services, and compute cost by ~30% through Graviton migration, multi-arch builds and Karpenter right-sizing.
- **Hardened platform security** with Kyverno admission policies and Vault dynamic secrets via Kubernetes and AWS STS auth; led CrowdStrike remediation with patched-AMI rollouts across 15+ AWS accounts.
- **Lead and mentor a 4-engineer team** supporting 5-6 application teams: set engineering standards, review design and code changes, and own GitLab CI, JFrog Artifactory with Vault, and the Docusaurus developer portal.

Conga Bangalore, India

Senior DevOps Engineer Apr 2024 – Jun 2025

- **Designed and built a SaaS API platform in Python and FastAPI** exposing 60+ REST endpoints, shipped through CI/CD with canary rollouts.
- **Architected a fault-tolerant, event-driven workflow orchestrator for CRM data migrations** in Python: Celery workers consume Kafka events, MongoDB holds workflow state, and a DAG engine resolves dependencies with automatic retries.
- **Scaled the orchestrator to 500+ concurrent workflows with zero data loss,** adding state tracking and a live monitoring dashboard, and raised migration automation coverage by 80%.
- **Led a 6-member DevOps/SRE team,** setting engineering standards, runbooks and SRE practices, and ran CI/CD knowledge-sharing sessions across engineering teams.

- **Cut deployment time from 25 minutes to under 8 across 200+ microservices,** and brought per-release downtime under 30 seconds with blue-green deployments.
- **Tripled deployment frequency and cut pipeline duplication 65%** across 8+ teams by building reusable Jenkins shared libraries and Spinnaker pipelines.
- **Reduced release failures 70% and rollback incidents 85%** by adding automated validation and auto-rollback, and fixing session-persistence and schema-migration failures during traffic shifts.
- **Designed event-driven autoscaling with KEDA** for NGINX on Prometheus request metrics and for AWS SQS consumers on queue depth, replacing fixed replica counts.
- **Made every deployment pass blocking quality gates:** Checkmarx security scans, compliance checks and Slack approvals built into the pipeline.

DevOps Engineer Apr 2023 – Mar 2024

- **Delivered disaster recovery for 200+ AWS services** with RTO under 10 minutes and RPO under 5 minutes through continuous replication and automated failover, with zero data loss across drills and real failovers.
- **Cut release cycles from 2 weeks to 3 hours** by automating highly available infrastructure with Terraform and adding artifact promotion, dynamic environments and rollback to pipelines.
- **Cut MTTR by 50%** by rolling out the observability stack (Prometheus, Grafana) and centralizing secrets in Doppler.
- **Containerized and migrated legacy Java and Python services to EKS** with autoscaling and health probes.
- **Hardened Terraform operations** with S3-backed remote state, locking and versioning, and wrote Groovy and shell tooling for build orchestration, pipeline linting and secret rotation.

Associate DevOps Engineer Jul 2021 – Mar 2023

- **Cut build times from 20 minutes to under 8** by parallelizing and caching Jenkins pipelines.
- **Cut new-service setup from 4 to 2.4 hours** with templatized Helm charts for stateless and stateful services, backed by automated chart validation and best-practice documentation.
- **Built foundational Terraform modules** for VPCs, IAM and EC2-based services across environments, supporting the move to cloud-native infrastructure.

TECHNICAL SKILLS

Languages: Python (FastAPI, Celery), Go, Bash, Groovy

Distributed Systems: Microservices, Event-driven Architecture, REST APIs, Kafka, Celery, Workflow Orchestration

Cloud: AWS (EKS, EC2, ALB, RDS, DynamoDB, S3, SQS, IAM, STS, Bedrock, Local Zones, Global Accelerator), GCP

Kubernetes: EKS, Docker, Custom CRDs and Controllers, Karpenter, KEDA, Helm, Argo CD, Kyverno

IaC and CI/CD: Pulumi, Terraform, GitLab CI, Jenkins, Spinnaker, Rancher Fleet, Blue-Green and Canary Releases

AI: Amazon Bedrock, Bedrock Knowledge Bases (RAG), Self-hosted SLMs on Kubernetes

Data Stores: MongoDB, DynamoDB, RDS, Redis, ScyllaDB, Aerospike

Observability and Security: Prometheus, Grafana, Loki, incident.io, HashiCorp Vault, Checkmarx, Doppler

EDUCATION

B.Tech, Computer Science and Engineering, M.S. Ramaiah University of Applied Sciences Jul 2017 – Jun 2021
CGPA 9.0

CERTIFICATIONS

- Certified Kubernetes Administrator (CKA)
- HashiCorp Certified: Terraform Associate